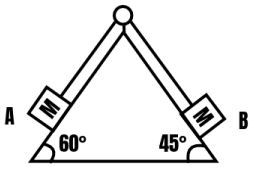
1 The dimensional formula for angular momentum is given by _____

A. ML^2T^2
B. ML^2T^{-1}
C. MLT^{-1}
D. MLT

2 Measurements for the outer and inner diameters of a hollow cylinder are (4.23 ± 0.01) cm and (3.89 ± 0.01) cm, respectively. What is the thickness of the cylinder's wall?

A. (0.34 ± 0.02) cm
B. (0.34 ± 0.01) cm
C. (0.17 ± 0.01) cm
D. (0.17 ± 0.02) cm

3 Two blocks, each possessing mass M, are placed on a frictionless double-inclined wedge as shown in the diagram. Under these conditions, _____



A. Block A slides down the incline
B. Block B slides down the incline
C. Both blocks remain motionless
D. Both blocks slide downwards

4 If the acceleration due to gravity at the Earth's surface is g, at what altitude above the surface will the value of gravity decrease to one-fourth of its surface value?

A. R
B. $\frac{R}{4}$
C. 4R
D. 2R

5 A satellite launched vertically from the Earth has an escape speed of 11 $\frac{km}{s}$. If the satellite is instead projected at an angle of 45° to the vertical, its escape velocity will be _____

A. 11 $\frac{km}{s}$
B. 22 $\frac{km}{s}$
C. 33 $\frac{km}{s}$
D. 44 $\frac{km}{s}$

6 As the temperature of a material increases, its Young's modulus of elasticity _____

A. increases
B. decreases
C. remains unchanged
D. drops to zero

7 When a liquid whose cohesive forces exceed its adhesive forces is placed in a glass container, the shape of the meniscus formed is _____

A. concave
B. convex
C. flat
D. oval

8 Compared to cold air, the coefficient of viscosity of hot air is _____

A. greater
B. smaller
C. identical
D. variable depending entirely on atmospheric pressure

9 A small spherical ball whose density is identical to that of a surrounding fluid is placed within it. The terminal velocity attained by the sphere is _____

A. 1
B. infinity
C. 100
D. 0

10 For a floating body to remain in stable equilibrium, which condition must be satisfied?

A. The metacenter lies below the center of gravity (C.G.)
B. The center of gravity (C.G.) lies below the metacenter
C. The metacenter coincides with the center of gravity (C.G.)
D. The metacentric height must equal zero

11 The thermal energy needed to raise the temperature of 1 g of water from 3.5°C to 4.5°C is _____

A. 1 Joule
B. 1 Calorie
C. 4.2 Calorie
D. 42 Calorie

12 Two distinct objects having a mass ratio of $\frac{3}{2}$ and a specific heat capacity ratio of $\frac{1}{4}$ absorb identical amounts of heat. The ratio of their temperature elevations is _____

A. $\frac{8}{3}$
B. $\frac{3}{8}$
C. $\frac{1}{4}$
D. $\frac{4}{1}$

13 At a temperature of 100°C, the saturated vapor pressure of water is _____

A. 766 mmHg
B. 716 mmHg
C. 760 mmHg
D. 750 mmHg

14 A closed container encloses 6 g of oxygen gas at a pressure P and temperature of 400 K. When gas escapes through a small pinhole, the final state reaches a pressure of $\frac{P}{2}$ at 300 K. The mass of oxygen that leaked out equals _____

A. 2 g
B. 3 g
C. 4 g
D. 5 g

15 According to Stefan's law, the rate of energy radiated by a blackbody at 300 K is directly proportional to _____

A. 300^1
B. 300^2
C. 300^4
D. 300

16 The current flowing through an electrical circuit drops uniformly from 5.0 A to 0.0 A in a span of 10 ms. Given a self-inductance of 10 mH, the magnitude of the induced electromotive force (emf) is _____

A. 50 V
B. 5 V
C. 1 V
D. 10 V

17 For a string clamped firmly at both ends, the frequencies of consecutive harmonics exist in the ratio _____

A. 1:2:3
B. 1:3:5
C. 1:1:1
D. 1:2:4

18 A steady current I passing through a metallic wire produces an electron drift velocity v. If both the electric current and the cross-sectional area of the conductor are doubled, the resulting drift velocity will be _____

A. 4v
B. $\frac{v}{2}$
C. $\frac{v}{4}$
D. v

19 Two acoustic waves of identical frequency propagate in a medium with sound intensity levels of 1 Bel and 5 Bel, respectively. The ratio of their amplitudes is _____

A. $\frac{1}{4}$
B. $\frac{1}{2}$
C. $\frac{1}{10^4}$
D. $\frac{1}{10^2}$

20 A sound source approaches a stationary listener, causing the observed pitch to be twice the emitted frequency. If the speed of sound in air is v, the speed of the approaching source is _____

A. v
B. $\frac{v}{2}$
C. 2v
D. $\frac{v}{4}$

21 The reddish coloration of the Sun observed during dusk and dawn is primarily caused by which optical phenomenon?

A. Reflection
B. Scattering
C. Interference
D. Total internal reflection

22 An equiconvex glass lens with a refractive index of 1.5 provides an optical power of +4 diopters. What is the radius of curvature of each refracting surface?

A. 25 cm
B. 100 cm
C. 4 cm
D. 10 cm

23 When two coherent waves with amplitudes 3a and a interfere constructively and destructively, the ratio of maximum intensity to minimum intensity is _____

A. $\frac{4}{1}$
B. $\frac{1}{4}$
C. $\frac{2}{1}$
D. $\frac{1}{2}$

24 The shimmering, vibrant coloration seen on a spiderweb illuminated by direct sunlight originates from _____

A. diffraction
B. refraction
C. reflection
D. interference

25 According to the inverse-square law, if the distance from a point light source to an illuminated surface is doubled, the illuminance _____

A. is halved
B. reduces to $\frac{1}{4}$
C. quadruples
D. doubles

26 A uniform electric field of strength E exists between the plates of a parallel plate capacitor separated by distance d with plate area A. The total electrostatic energy stored inside is _____

A. $\frac{1}{2}\epsilon_0 E^2$
B. $\frac{1}{2}\epsilon_0 E^2 Ad$
C. $\frac{1}{2}EAd$
D. $\frac{E^2 Ad}{\epsilon_0}$

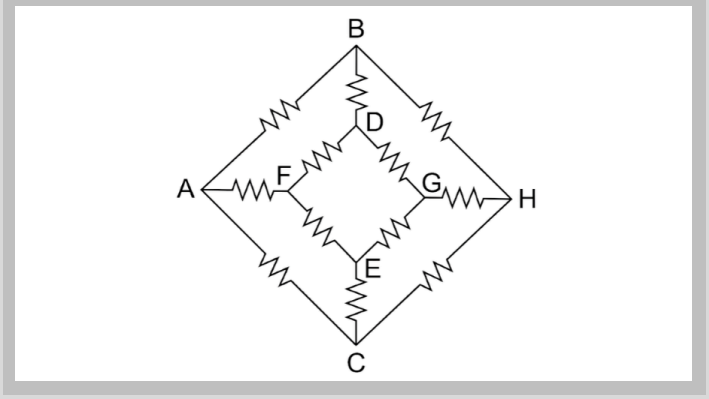
27 A group of n identical capacitors are first coupled in series and subsequently arranged in parallel. The ratio between the maximum and minimum equivalent capacitances obtained is _____

A. $\frac{1}{n^2}$
B. $\frac{1}{n}$
C. n
D. n^2

28 A metallic conductor of length 50 cm and cross-sectional area 1 mm^2 conducts a current of 4 A when energized by a 2 V source. The electrical resistivity of the material is _____

A. $4 \times 10^{-6}\ \Omega$ m
B. $2 \times 10^{-7}\ \Omega$ m
C. $5 \times 10^{-7}\ \Omega$ m
D. $1 \times 10^{-6}\ \Omega$ m

29 In the symmetrical resistor network shown, determine the equivalent resistance between nodes A and H, given that each branch resistor equals 1 Ω.



A. 1 Ω
B. $\frac{5}{3}\Omega$
C. $\frac{3}{4}\Omega$
D. $\frac{7}{6}\Omega$

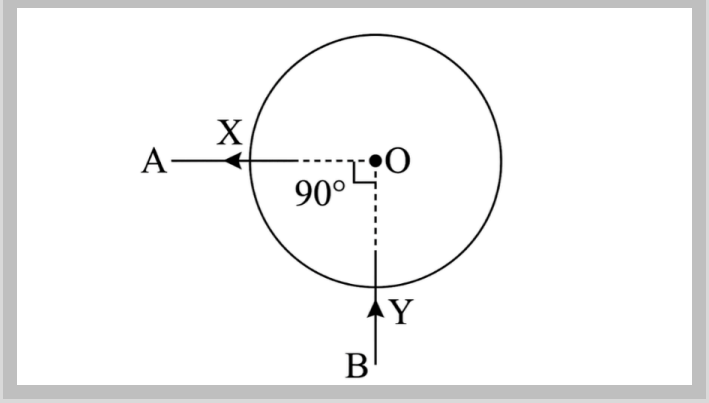
30 A battery possesses an emf of 6 V alongside an internal resistance of 0.5 kΩ. When connected across a voltmeter of internal resistance 2.5 kΩ, the observed terminal voltage reads _____

A. 1 V
B. 10 V
C. 5 V
D. 10^{-3} V

31 In thermoelectricity, if θ_n denotes the neutral temperature, θ_i the inversion temperature, and θ_0 the cold junction temperature, which formula expresses their relationship?

A. $2\theta_n = \theta_0 + \theta_i$
B. $2\theta_n = \theta_0 - \theta_i$
C. $\theta_n = \theta_0 + \theta_i$
D. $\theta_i = \theta_0 + \theta_i$

32 For a current-carrying wire bent into a three-quarter circular loop of radius r carrying steady current I as shown, the magnetic field at the center O is _____



A. $\frac{\mu_0 I}{4r}$
B. $\frac{2\mu_0 I}{3r}$
C. $\frac{3\mu_0 I}{4r}$
D. $\frac{3\mu_0 I}{8r}$

33 A circular conducting loop carrying an electric current is suspended horizontally so that it can rotate freely. In equilibrium with Earth's magnetic field, the plane of the loop aligns _____

A. in the North–South direction
B. in the East–West direction
C. in whichever direction it is released
D. at an angle of 45° to the East–West direction

34 Although a proton is roughly 1836 times more massive than an electron, accelerating it across an electric potential difference of 1 V imparts a kinetic energy of _____

A. 1836 eV
B. 1 eV
C. 1 MeV
D. 1836 MeV

35 An energy value of 1 electron-volt (1 eV) corresponds precisely to _____

A. 1.6×10^{-19} J
B. 10 erg
C. 1.1×10^{-38} erg
D. 1.1×10^{-11} erg

36 In a photoelectric experiment, the value of the saturation photocurrent is governed primarily by the _____

A. intensity of incident light
B. stopping potential
C. frequency of radiation
D. wavelength of photons

37 When plotting a graph of stopping potential versus the frequency of incident radiation for the photoelectric effect, the slope of the resulting straight line yields the _____

A. kinetic energy of emitted electrons
B. work function of the emitter metal
C. ratio of Planck's constant to electronic charge
D. saturation photoelectric current

38 Provided that an electron and a proton share the exact same de Broglie wavelength, which particle possesses the greater kinetic energy?

A. Electron
B. Proton
C. Both have the same kinetic energy
D. Insufficient information to determine

39 In comparison with soft X-rays, hard X-rays are characterized by having _____

A. shorter wavelength
B. lower energy
C. lower frequency
D. longer wavelength

40 If the mass numbers of two atomic nuclei are in the proportion of $\frac{1}{3}$, the ratio of their respective nuclear densities is _____

A. $\frac{3^{\frac{1}{3}}}{1}$
B. $\frac{1}{3}$
C. $\frac{3}{1}$
D. $\frac{1}{1}$

41 The refractive index of an optical medium is entirely independent of the _____

A. temperature
B. intensity of incident light
C. wavelength of light
D. optical density

42 Identify the unknown particles X and Y in the nuclear transformations: ${}_7N^{14} + {}_0n^1 \rightarrow {}_6C^{14} + X$ and ${}_5B^{10} + {}_2He^4 \rightarrow {}_7N^{13} + Y$.

A. ${}_1H^1$ and ${}_0n^1$
B. ${}_0n^1$ and ${}_{+1}e^0$
C. ${}_{-1}e^0$ and ${}_1H^1$
D. ${}_0n^1$ and ${}_1H^1$

43 To fabricate a p-type extrinsic semiconductor, a pure intrinsic semiconductor crystal is doped with which class of dopant atoms?

A. Tetravalent
B. Pentavalent
C. Trivalent
D. Divalent

44 The formation of a potential barrier across an unbiased p–n junction diode is fundamentally caused by the _____

A. diffusion of majority charge carriers
B. drift of free electrons
C. drift of mobile holes
D. energy of the forbidden band

45 A standard center-tapped full-wave rectifier circuit requires a minimum of how many p–n junction diodes?

A. 0 B. 1
C. 2 D. 3

46 In a transistor operating in common-base mode, the emitter current is measured at 5.60 mA while the collector current is 5.488 mA. What is the common-emitter current amplification factor (β)?

A. 49 B. 48
C. 50 D. 51

47 Which valence quark combination represents an antineutron?

A. udd B. uud
C. $\bar{u}\bar{d}\bar{d}$ D. $\bar{u}ud$

48 Galaxy B is situated at double the distance from Earth compared to galaxy A. According to Hubble's law, the recessional speed of galaxy B relative to Earth is approximately _____

A. 2 times the velocity of galaxy A
B. 4 times the velocity of galaxy A
C. equal to the velocity of galaxy A
D. zero velocity

49 According to standard cosmological estimates, what percentage of the total energy-mass content of the universe consists of dark matter?

A. 85% B. 25%
C. 15% D. 30%

50 An alternating source with an rms potential of 200 V at 50 Hz feeds a series LCR network. A voltage drop exceeding 200 V may develop across _____

A. the resistor
B. the inductor only
C. the capacitor only
D. either the inductor or the capacitor depending on the L and C values

51 Indigo dye is classified chemically as which variety of coloring agent?

A. Vat dye B. Phthalein indicator
C. Basic dye D. Mordant dye

52 Which of the following substances functions as a standard reference electrode in electrochemistry?

A. $CuSO_4$ B. Corrosive sublimate
C. Zinc oxide D. Calomel

53 Tetrafluoroethylene, $F_2C = CF_2$, undergoes addition polymerization to yield _____

A. TFE B. Nylon
C. Teflon D. Polyethene

54 While lemons are predominantly rich in citric acid, apples primarily contain _____

A. Lactic acid B. Malic acid
C. Citric acid D. Tartaric acid

55 Among the following substituted phenols, which demonstrates the highest acidic strength?

A. p-nitrophenol B. o-nitrophenol
C. m-nitrophenol D. o-aminophenol

56 Which alkali metal possesses the unique ability to react directly with atmospheric nitrogen upon heating to form an ionic nitride?

A. Na B. Li
C. Cs D. K

57 The crude, high-carbon molten iron extracted straight from a blast furnace is designated as _____

A. Wrought iron B. Pig iron
C. Mild iron D. Cast iron

58 An inorganic compound X possesses a pungent, rotten-fish odor and produces a black precipitate when exposed to aqueous Cu^{2+} ions. Compound X is _____

A. H_2S B. NH_3
C. CH_4 D. PH_3

59 Which allotropic modification of carbon can exhibit superconductivity when suitably doped with alkali metals?

A. Fullerene B. Graphite
C. Diamond D. Lamp black

60 In a molecule where a carbon atom undergoes sp^2 hybridization, it is directly covalently bonded to how many surrounding atoms?

A. 2 B. 3
C. 4 D. 5

61 The chemical extraction or removal of oxygen from a compound is formally defined as _____

A. Reduction B. Oxidation
C. Hydration D. Dehydrogenation

62 **Nitrobenzene is widely known by which historical commercial name?**

A. Oil of mirbane
B. Oil of vitriol
C. Oil of bitter almond
D. Oil of cinnamon

63 **Controlled catalytic hydrogenation of an internal alkyne in the presence of Lindlar's catalyst selectively yields a(n) _____**

A. terminal alkene
B. cis-alkene
C. trans-alkene
D. alkane

64 **The nucleophilic addition of methyl magnesium halide to acetonitrile, $CH_3MgX + CH_3CN$, followed by acidic hydrolysis, generates _____**

A. Acetone
B. Ethanol
C. Acetaldehyde
D. Ethanoic acid

65 **When warmed with iodine in the presence of an alkali, which compound gives a positive iodoform test with a characteristic antiseptic odor?**

A. 2-pentanone
B. Propanal
C. 3-pentanone
D. NH_3

66 **Which of the following carboxylic acids has the smallest pK_a value (greatest acidity)?**

A. CH_3COOH
B. HCOOH
C. CH_3CH_2COOH
D. $CH_3CH_2CH_2COOH$

67 **What is the correct sequence of ionic and atomic radii for the iodine species?**

A. \(\text{I} > \text{I}^+ > \text{I}^-\
B. $I^- > I > I^+$
C. $I^- > I^+ > I$
D. $I > I^- > I^+$

68 **Which of the following chemicals is not utilized in the industrial bleaching of wood pulp?**

A. $CaOCl_2$
B. NH_3
C. Cl_2
D. NaOH

69 **The charring of glucose by concentrated sulfuric acid, represented by $C_6H_{12}O_6 + \text{conc. } H_2SO_4 \rightarrow 6C + 6H_2O \cdot H_2SO_4$, is an example of a _____**

A. disproportionation reaction
B. dehydration reaction
C. displacement reaction
D. substitution reaction

70 **Which of the following transformations does not accurately depict a reaction involving a Grignard reagent?**

A. Ester reacting to give a tertiary alcohol
B. Aldehyde reacting to give an alcohol
C. Acetaldehyde reacting to give a tertiary alcohol
D. CO_2 reacting to give a carboxylic acid

71 **The acceleration of ester hydrolysis in the presence of hydrogen ions (H^+) illustrates which type of catalysis?**

A. Heterogeneous catalysis
B. Autocatalysis
C. Catalytic promotion
D. Negative catalysis

72 **Kinetic analysis of a chemical reaction yields the following data: (i) [P] = 0.16, [Q_2] = 0.16, Rate = 0.16; (ii) [P] = 0.16, [Q_2] = 0.32, Rate = 0.32; (iii) [P] = 0.32, [Q_2] = 0.32, Rate = 0.32. The corresponding rate expression is _____**

A. $[R] = [P][Q_2]$
B. $[R] = [Q_2]$
C. $[R] = [Q_2]^2$
D. $[R] = [P][Q_2]^2$

73 **What is the pH of a 0.02 N aqueous solution of sodium hydroxide?**

A. 12.7
B. 13.0
C. 11.5
D. 14.0

74 **If an aqueous solution contains 0.1 $\frac{g}{L}$ of dissolved arsenic, its concentration in parts per million (ppm) is _____**

A. 1000 ppm
B. 1 ppm
C. 100 ppm
D. 10 ppm

75 **The reduction of sodium iodate solution using sodium bisulfite predominantly produces _____**

A. Iodine
B. Sulfur
C. SO_2
D. HI

76 **Ice floats on water because its lower density arises from _____**

A. an open, cage-like lattice maintained by intermolecular hydrogen bonding
B. an open, cage-like structure formed by intramolecular hydrogen bonds
C. strong directional covalent bonding
D. extensive van der Waals attractions

77 **In the Perkin reaction, identify reactant X: X + CH_3COONa + $(CH_3COO)_2O \rightarrow$ Cinnamic acid.**

A. C_6H_5CHO
B. C_6H_5COOH
C. $C_6H_5NO_2$
D. $C_6H_5NH_2$

78 **When an α-amino acid undergoes enzymatic or chemical decarboxylation, the organic product formed is a(n) _____**

A. carboxylic acid
B. alcohol
C. primary amine
D. ethanol

79 **A monocarboxylic acid has an empirical formula of CH_2O_2. Its molecular formula is _____**

A. CH_2O_2
B. $C_2H_4O_4$
C. CH_2O
D. $C_2H_2O_3$

80 **A binary chemical compound contains 60% of element X (atomic mass 30) and 40% of element Y (atomic mass 13) by mass. What is its empirical formula?**

A. X_2Y_3 B. X_3Y_2
C. XY_2 D. X_3Y

81 **A chemist recognizes that hydrogen sulfide, whether produced in the human digestive tract or synthesized in a laboratory, has identical chemical composition. This conforms to the _____**

A. Law of Conservation of Mass
B. Law of Multiple Proportions
C. Law of Definite Proportions
D. Law of Reciprocal Proportions

82 **Which of the following substances has the highest liquid density at standard conditions?**

A. Hg B. H_2O
C. Glycerol D. NH_3

83 **Among the following non-metals, which displays the greatest tendency toward catenation?**

A. S B. C
C. Si D. O

84 **When phenol is heated with chloroform and aqueous sodium hydroxide (Reimer–Tiemann reaction), the principal product obtained is _____**

A. Salicylic acid B. Salicylaldehyde
C. Benzoic acid D. Phenyl carbylamine

85 **A pharmaceutical drug administered to lower an elevated body temperature during fever is termed a(n) _____**

A. Analgesic B. Antiseptic
C. Antipyretic D. Disinfectant

86 **Which organic compound reduces Tollens' reagent to precipitate an elemental silver mirror?**

A. Acetic acid B. Acetone
C. Ethanol D. Acetaldehyde

87 **Which of the following tetrahedral molecules has a net dipole moment of zero due to symmetry?**

A. CCl_4 B. $CHCl_3$
C. CH_3Cl D. CH_2Cl_2

88 **The classical Bohr model of the atom proved inadequate in explaining the _____**

A. hydrogen emission spectrum
B. Stark and Zeeman effects
C. line spectrum of single-electron ions
D. quantization of angular momentum as an integral multiple of $\frac{h}{2\pi}$

89 **The oxidation state of the central sulfur atom in sulfuric acid, H_2SO_4, is _____**

A. +2 B. +6
C. +5 D. +3

90 **Which allotropic form of phosphorus glows in the dark due to slow oxidation (phosphorescence)?**

A. Scarlet phosphorus B. Red phosphorus
C. Black phosphorus D. White phosphorus

91 **Which of the following neutral or ionic species shares the same number of electrons as the potassium ion (K^+)?**

A. S B. Na^+
C. Ar D. Ca^{++}

92 **Which of the following substances does not serve as a reducing agent in extractive metallurgical processes?**

A. CO B. CO_2
C. H_2 D. C

93 **Which element ignites in air to yield an oxide that remains in the gaseous state at room temperature?**

A. Na B. H
C. S D. He

94 **During radioactive decay, which type of emission leaves the atomic number of the daughter nucleus unaltered?**

A. γ emission
B. β emission
C. α emission
D. Electron capture/beta emission

95 **A person entered an abandoned deep well to clean it and died shortly after from asphyxiation. Which dense gas accumulating at the bottom was most likely responsible?**

A. CO_2 B. CO
C. SO_2 D. HCN

96 **Which essential plant macronutrient is particularly critical for stimulating floral development and seed formation?**

A. Phosphorus B. Nitrogen
C. Potassium D. Calcium

97 **Which chemical species possesses a nuclear charge of 11 protons alongside 10 extranuclear electrons?**

A. Ne B. Na
C. Na^+ D. Na^-

98 **Which thermodynamic and physical condition predominantly stabilizes matter in the solid state?**

A. High thermal energy B. Low temperature
C. High temperature D. Weak cohesive forces

99 **Which class of organic compounds innately possesses two distinct functional groups in its monomeric form?**

A. Carboxylic acid
B. Acid anhydride
C. Ester
D. Amino acid

100 **Which of the following elements, being chemically unreactive and non-toxic to human physiology, is harmless when inhaled in minor amounts?**

A. Fe
B. Cu
C. Hg
D. Ar

101 **The polymerase chain reaction (PCR) method was originally devised in 1983 by _____**

A. Kary Mullis
B. Sidney Altman
C. César Milstein
D. Georges Köhler

102 **An interspecific biotic interaction wherein both participating populations experience mutual detriment is known as _____**

A. Competition
B. Amensalism
C. Predation
D. Parasitism

103 **A homozygous mother with blood type B (** Double exponent: use braces to clarify **) conceives with a father of blood type A. What are the potential phenotypes for their offspring?**

A. A and AB
B. AB and B
C. O and B
D. O and AB

104 **The biological discipline devoted to the structure, function, and interactions of nucleic acids like DNA and RNA is _____**

A. Genetics
B. Physiology
C. Cytology
D. Molecular biology

105 **The accidental loss of an allele from a small, isolated gene pool due to the random death or departure of its sole carrier represents an example of _____**

A. Gene pool
B. Genetic drift
C. Mutation
D. Ecological disaster

106 **Which monohybrid cross produces offspring exhibiting an exact $\frac{1}{1}$ genotypic proportion?**

A. TT and tt
B. tt and Tt
C. Tt and Tt
D. Tt and TT

107 **During nucleic acid biosynthesis, DNA polymerase catalyzes the elongation of a daughter strand in the _____**

A. 5' to 3' direction of DNA
B. 3' to 5' direction of DNA
C. 5' to 3' direction of m-RNA
D. 3' to 5' direction of m-RNA

108 **The terminal chromosomal segment situated distal to the secondary constriction is designated as the _____**

A. Telomere
B. Satellite (trabant)
C. Centromere
D. Nucleolar organizer region (NOR)

109 **The protoplasmic theory, which stated that the essential physical basis and living substance of a cell is protoplasm, was formulated by _____**

A. Max Schultze
B. Hugo von Mohl
C. Félix Dujardin
D. Jan Evangelista Purkinje

110 **The opening movements of floral petals and downward curling of growing buds are classic instances of _____**

A. Epinasty
B. Hyponasty
C. Chemotaxis
D. Thigmonasty

111 **The gain of a single extra chromosome into a normal diploid complement (2n + 1) is referred to as _____**

A. Trisomy
B. Nullisomy
C. Monosomy
D. Tetrasomy

112 **Which genetic disorder results directly from the presence of a third copy of chromosome 18 (trisomy 18)?**

A. Patau syndrome
B. Down syndrome
C. Edwards syndrome
D. Klinefelter syndrome

113 **The occurrence of three or more alternative allelic variants of a single gene governing a specific trait within a population is termed _____**

A. Pleiotropy
B. Polygene inheritance
C. Multiple allelism
D. Incomplete dominance

114 **How many genetically distinct gametic types can an individual with the genotype AaBbDdEE produce through independent assortment?**

A. 4
B. 8
C. 16
D. 2

115 **When cells are incubated with carbon-14 labeled uracil (^{14}C-uracil), which biomacromolecule specifically incorporates the radioactivity?**

A. Protein
B. DNA
C. Monosaccharide
D. RNA

116 **The siliqua, a dry dehiscent fruit characteristic of Brassicaceae, originates from a(n) _____**

A. monocarpellary ovary
B. bicarpellary ovary
C. tricarpellary ovary
D. multicarpellary ovary

117 **Vascular bundles that are conjoint, collateral, and open (with active intrafascicular cambium) are characteristic of the _____**

A. dicot stem
B. monocot stem
C. dicot root
D. monocot root

118 **In plant tissue culture, surface sterilization of an explant prior to inoculation is routinely performed using _____**

A. membrane filtration with pore diameter under 0.2 microns
B. prolonged exposure to UV radiation
C. vacuum distillation
D. sodium hypochlorite solution

119 **Commercial bast fibres of jute are extracted from bark. Anatomically, bark encompasses _____**

A. all tissues exterior to the vascular cambium, containing secondary phloem fibres
B. all tissues exterior to the phellogen, comprising only phellem
C. all cellular layers located outside the pith
D. only tissues positioned exterior to the endodermis

120 **Which biological vector system is most widely utilized for gene transfer and transformation in plant genetic engineering?**

A. Agrobacterium tumefaciens-mediated gene transfer
B. CRISPR-Cas9 genome editing
C. Micropropagation
D. Callus tissue culture

121 **The wavelengths of the electromagnetic spectrum that effectively drive the photochemical action spectrum of photosynthesis span primarily between _____**

A. white to yellow
B. red to green
C. red to yellow
D. white to green

122 **The deliberate alteration and in vitro recombination of an organism's genetic material to confer novel beneficial traits, such as disease resistance, is defined as _____**

A. Genetic engineering
B. Biotechnology
C. Gene pooling
D. Gene mutation

123 **Sal (Shorea robusta) and teak (Tectona grandis) constitute the predominant climax tree species in _____**

A. tropical dry deciduous forests
B. temperate rainforests
C. temperate broadleaf forests
D. needleleaf coniferous forests

124 **Identify the incorrect etymological pair among the following botanical terms:**

A. Phanerogams — Visible reproduction
B. Bryon — Liverworts
C. Gymnos — Naked seeds
D. Kryptos — Hidden or concealed

125 **A complex ecological network formed by the interlacing and cross-connection of multiple food chains is termed a _____**

A. complex food chain
B. trophic cycle
C. food web
D. ecological biomass pyramid

126 **In typical angiosperms, the primary endosperm tissue formed via double fertilization is _____**

A. haploid (n)
B. diploid (2n)
C. triploid (3n)
D. pentaploid (5n)

127 **In horticulture, the technique of grafting is predominantly valued because _____**

A. it produces vast numbers of seed offspring
B. it is the simplest and cheapest multiplication method
C. it combines the desired traits of both scion and rootstock
D. it generates genetically novel hybrid progeny

128 **Which statement accurately describes archaebacteria?**

A. All members are obligate photoautotrophs
B. All known representatives exist only as fossils
C. They represent some of the oldest extant living lineages
D. All species are strictly halophilic

129 **Internodal elongation and premature flowering (bolting) in rosette plants is primarily induced by the application of _____**

A. Gibberellins
B. Auxins
C. Cytokinins
D. Abscisic acid (ABA)

130 **Mitochondria are regarded as semi-autonomous cellular organelles primarily because they possess _____**

A. oxysome complexes
B. their own circular DNA and protein-synthesizing machinery
C. internal cristae and mitochondrial matrix
D. no cellular RNA

131 **Which of the following gaseous emissions represents the major primary atmospheric pollutant?**

A. SO_2
B. NO_2
C. Lead particulates
D. CO

132 **The tremendous biological diversity characteristic of the Terai belt of Nepal is primarily attributable to its _____**

A. soil pH
B. prevailing warm temperature
C. annual precipitation patterns
D. low elevation and altitudinal gradient

133 **Which of the following statements concerning the production of beer and wine is accurate?**

A. Yeast ferments beer while bacteria exclusively ferment wine
B. Beer is derived from fermented malt, and wine is produced from fermented molasses
C. Beer is brewed from fermented malted cereals, whereas wine is crafted from fermented fruit juice
D. Both beverages are derived strictly from malted barley

134 **The artificial system of plant taxonomy termed the 'sexual system' based on stamens and carpels was proposed by _____**

A. Aristotle
B. Theophrastus
C. Carolus Linnaeus
D. Ernst Haeckel

135 **In the taxonomic hierarchy, organisms grouped within which category share the lowest degree of mutual similarity?**

A. Class
B. Order
C. Family
D. Genus

136 **The long-term conservation of viable biological germplasm or tissues at ultra-low temperatures (often in liquid nitrogen at -196°C) is known as _____**

A. Cold storage
B. Cryopreservation
C. Cold chain maintenance
D. Lyophilization vialing

137 **A bisexual floral structure that never opens at anthesis and enforces self-pollination within the bud is described as a _____**

A. cleistogamous flower
B. dichogamous flower
C. homogamous flower
D. heterogamous flower

138 **In plant ecological succession, the term 'ecesis' refers specifically to the _____**

A. successful germination and establishment of a colonizing species in a new habitat
B. attainment of a climax community
C. steady-state environmental equilibrium
D. initial migration and influx of propagules

139 **Which grouping comprises exclusively vascular plants (tracheophytes) possessing specialized xylem and phloem?**

A. Bryophytes, Pteridophytes, Gymnosperms, and Angiosperms
B. Pteridophytes, Gymnosperms, and Angiosperms
C. Gymnosperms and Pteridophytes only
D. Gymnosperms and Angiosperms only

140 **In sugarcane and maize, stilt roots (prop roots providing mechanical anchoring) develop adventitiously from the _____**

A. lower internodes
B. upper internodes
C. lower basal nodes
D. terminal stem nodes

141 **Which anatomical feature accurately characterizes the nematode Ascaris lumbricoides?**

A. Absence of both a true coelom and true metamerism
B. Absence of a true coelom with presence of metamerism
C. Presence of both a true coelom and metamerism
D. Presence of a true coelom with absence of metamerism

142 **While surveying a marine coastline, a biologist discovers a pearl-bearing bivalve mollusc. This organism belongs to the genus _____**

A. Physalia
B. Pinctada
C. Mytilus
D. Dentalium

143 **In Pheretima posthuma, the terminal excretory ducts of the septal nephridia discharge their contents into the _____**

A. lumen of the intestine
B. paired septal excretory canal
C. paired supra-intestinal excretory duct
D. dorsal blood vessel

144 **A digitigrade locomotor posture, where an animal walks on its digits with heels elevated, is primarily an adaptation for _____**

A. fossorial burrowing
B. arboreal climbing
C. cursorial running
D. volant flight

145 **Just as iron is central to hemoglobin and magnesium to chlorophyll, which metallic trace element forms the core of cyanocobalamin (vitamin B_{12})?**

A. Cobalt
B. Iron
C. Magnesium
D. Sulfur

146 **In the mammalian heart, the Eustachian valve guards the opening of the _____**

A. superior vena cava
B. inferior vena cava
C. pulmonary vein
D. systemic aorta

147 **Which mammalian subclass or group naturally lacks an external ear pinna?**

A. Prosimians
B. Prototheria (monotremes)
C. Eutheria (placentals)
D. Metatheria (marsupials)

148 **Acute mountain sickness experienced by travelers ascending to high altitudes is primarily triggered by the _____**

A. reduced partial pressure of oxygen
B. elevated carbon dioxide concentration
C. systemic dehydration
D. diminished intrinsic binding affinity of hemoglobin

149 **During physiological acclimatization to high-altitude environments over several days, the human body responds by _____**

A. stimulating erythropoiesis to increase RBC count
B. suppressing red blood cell production
C. increasing the physical dimensions of individual RBCs
D. decreasing total hemoglobin concentration

150 **Which type of lingual papillae is rudimentary or entirely absent on the adult human tongue?**

A. Circumvallate papillae
B. Filiform papillae
C. Fungiform papillae
D. Foliate papillae

151 **In the digestive anatomy of the frog, where is the gallbladder situated relative to the liver lobes?**

A. Between the two small left lobes of the liver
B. Between the right lobe and left lobe
C. Embedded within liver tissue and rendered invisible
D. Between the subdivisions of the right lobe

152 **Which human organ acts as the main reservoir for Plasmodium sporozoites during the initial pre-erythrocytic developmental phase?**

A. Liver
B. Kidney
C. Jejunum
D. Spleen

153 **Areolar loose connective tissue is extensively distributed between the _____**

A. skin integument and underlying skeletal muscles
B. bones and adipose bodies
C. bones and attached muscles
D. skeletal muscles and visceral fat

154 **The human heart is classified as myogenic because its intrinsic rhythmicity and contraction impulses originate from the _____**

A. bundle of His
B. Purkinje network
C. sinoatrial (SA) node
D. atrioventricular (AV) node

155 **The central neural thermostat regulating mammalian body temperature is situated in the _____**

A. Hypothalamus
B. Diencephalon roof
C. Cerebellum
D. Cerebrum cortex

156 **Homeothermic endotherms maintain a stable body temperature through _____**

A. conforming their internal temperature passively to surrounding shifts
B. sustaining a constant internal thermal homeostasis despite external fluctuations
C. continuously elevating metabolic heat generation regardless of climate
D. actively suppressing metabolic rate in cold weather

157 **Which of the following psychoactive drugs acts pharmacologically as a central nervous system depressant?**

A. Amphetamines
B. LSD
C. Barbiturates
D. Cocaine

158 **Which primary animal tissue lines internal cavities, covers external body surfaces, and provides protective barriers?**

A. Epithelial tissue
B. Connective tissue
C. Nervous tissue
D. Muscular tissue

159 **According to the biochemical theory of the origin of life, the earliest primitive living cells on Earth were _____**

A. cyanobacteria
B. autotrophs
C. photoautotrophs
D. chemoheterotrophs

160 **According to Oparin and Haldane's abiogenesis model, which self-assembling prebiotic colloidal aggregations gave rise to primitive chemoheterotrophs?**

A. Free amino acids
B. Coacervates
C. Chlorophyll complexes
D. Polysaccharide carbohydrates

161 **In a starved specimen of Hydra, which gastrodermal cell type exhibits intense vacuolation due to intracellular food digestion?**

A. Epitheliomuscular cells
B. Nutritive-muscular cells
C. Interstitial cells
D. Glandular cells

162 **Which characteristic universally describes organisms classified under the subkingdom Protozoa?**

A. They are unicellular organisms
B. They contain only a single vacuole
C. They have only one nucleus
D. They possess a rigid cellulose cell wall

163 **The amplified croaking vocalizations produced by male frogs during the mating season are facilitated by _____**

A. high-frequency vibrating vocal cords
B. air-inflated vocal sacs functioning as acoustic resonators
C. two distinct pairs of vocal pouches
D. highly hypertrophied laryngeal muscles

164 **Human immunodeficiency virus (HIV) can be viably transmitted through which of the following routes?**

A. Social hugging
B. Casual kissing
C. Ingesting contaminated food or water
D. Ingestion of infected breast milk

165 **Body symmetry in which an organism can be partitioned into two equivalent halves along any longitudinal plane intersecting the central axis is designated as _____**

A. Biradial symmetry
B. Radial symmetry
C. Bilateral symmetry
D. Asymmetry

166 **Glomerular ultrafiltration in the nephron occurs because the _____**

A. glomerular hydrostatic pressure exceeds the combined blood colloidal osmotic and capsular pressures
B. blood colloidal osmotic pressure surpasses the glomerular hydrostatic pressure
C. capsular hydrostatic pressure and oncotic pressure together exceed glomerular pressure
D. capsular hydrostatic pressure is greater than the glomerular capillary pressure

167 **According to IUCN criteria, a taxon facing a very high risk of extinction in the wild in the near future if detrimental factors persist is classified as _____**

A. Threatened
B. Vulnerable
C. Rare
D. Endangered

168 **In human infants, which proteolytic gastric enzyme coagulates soluble casein into insoluble paracasein for milk digestion?**

A. Chymotrypsin
B. Rennin (chymosin)
C. Pepsin
D. Lactase

169 **Which ovarian and luteal hormone plays the principal role in preparing the endometrium for successful blastocyst implantation?**

A. Follicle-stimulating hormone (FSH)
B. Estrogen
C. Progesterone
D. Prolactin

170 **The electrical potential across a neuronal membrane maintained in its polarized state prior to stimulation is known as the _____**

A. action potential
B. resting membrane potential
C. threshold potential
D. hyperpolarization potential

171 **A patient presenting with central nervous system sedation, pronounced muscular relaxation, tranquilization, and somnolence has likely abused _____**

A. Valium (diazepam)
B. LSD
C. Cocaine
D. Peyote

172 **Which specialized epithelial tissue lines the urinary bladder and readily distends under hydrostatic tension?**

A. Squamous epithelium
B. Columnar epithelium
C. Cuboidal epithelium
D. Transitional epithelium (urothelium)

173 **Parathyroid hormone (PTH) functions biochemically in calcium and phosphate homeostasis by _____**

A. regulating cerebral calcium and phosphorus storage
B. increasing calcium concentrations while decreasing phosphate levels in blood plasma
C. decreasing circulating calcium and depositing phosphate in bone matrix
D. reducing both calcium and phosphate concentrations in systemic circulation

174 **Wading or bathing in contaminated freshwater exposes individuals to infection when free-swimming cercariae larvae of which parasite bore directly through intact skin?**

A. Ancylostoma duodenale
B. Schistosoma mansoni
C. Wuchereria bancrofti
D. Trypanosoma brucei

175 **The skeleton of the laryngotracheal chamber in Rana tigrina is constructed from which configuration of cartilages?**

A. 1 arytenoid and 2 cricoids
B. 2 arytenoids and 1 cricoid
C. 1 cricoid and 1 arytenoid
D. 2 cricoids and 2 arytenoids

176 The chloragogen cells surrounding the annelid intestine perform deamination and glycogen storage, making them functionally analogous to the vertebrate _____

A. kidney
B. liver
C. intestinal mucosa
D. lungs

177 The amphibian heart is anatomically distinguished from the mammalian heart by possessing a distinct auxiliary chamber termed the _____

A. sinus venosus
B. tricuspid valve
C. bicuspid (mitral) valve
D. pair of completely partitioned atria

178 Wharton's duct serves as the principal excretory drainage conduit for which salivary gland?

A. Submandibular (submaxillary) gland
B. Parotid gland
C. Sublingual gland
D. Infraorbital gland

179 The unexpected genetic expression of a primitive ancestral phenotypic trait in a modern organism is referred to as _____

A. analogy
B. homology
C. vestigiality
D. atavism

180 All of the following represent established clinical modalities for cancer management except _____

A. surgical excision
B. chemotherapy
C. routine physiotherapy
D. radiotherapy

181 Under a specific numerical coding rule where the letters of consonants and vowels are evaluated such that $\text{KIDNEY} = \frac{54}{14}$, the equivalent code for BRAIN is _____

A. $\frac{34}{10}$
B. $\frac{32}{22}$
C. $\frac{54}{22}$
D. $\frac{61}{10}$

182 Complete the numerical sequence by identifying the missing value: 2, 12, 30, 56, ?, 132.

A. 49
B. 90
C. 86
D. 102

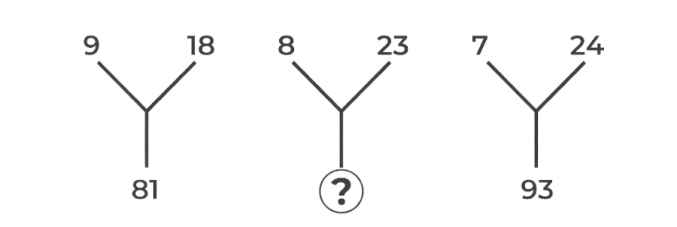
183 Purchasing 2 bus tickets to city B and 3 to city C from terminal A amounts to $77, whereas 3 tickets to B and 2 to C cost $73. What are the respective individual fares to destination B and destination C?

A. $4 and $23
B. $13 and $17
C. $15 and $14
D. $17 and $13

184 An examination comprises 500 objective questions. A student correctly answers 80% of the first 250 questions. What percentage of the remaining 250 questions must he answer correctly to attain a cumulative score of 60%?

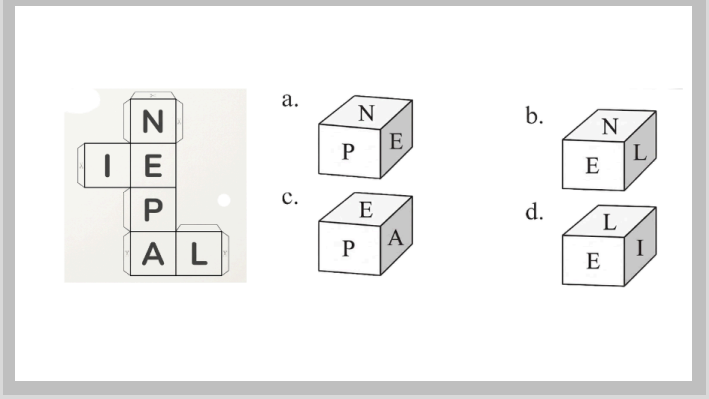
A. 30%
B. 40%
C. 50%
D. 60%

185 Determine the missing term denoted by '?'.



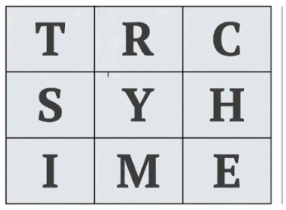
A. 45
B. 56
C. 72
D. 93

186 Which 3D cube is formed when the given net is assembled?



A. Box a
B. Box b
C. Box c
D. Box d

187 A valid English word is created starting from one corner letter and traversing strictly through orthogonally adjacent letters. What is the fifth letter in this nine-letter word?



T	R	C
S	Y	H
I	M	E

A. Y
B. E
C. I
D. S

188 Identify the missing value denoted by '?'.

3	9	5	E
4	4	1	H
21	39	18	?

A. 21
B. 12
C. 26
D. 39

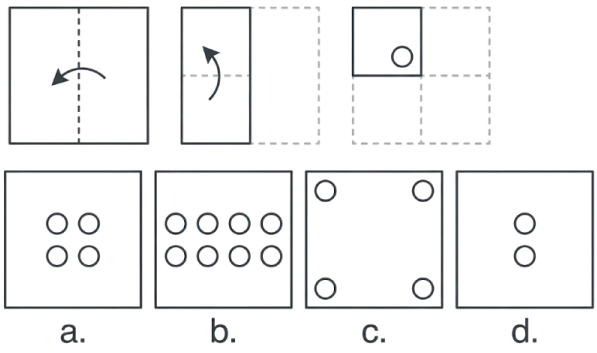
189 A faculty college organizes a 7-day refresher program in six subjects (Microbiology, Physiology, Surgery, Anatomy, Biochemistry, Orthopaedics) from July 22 to July 28: (i) Physiology begins the program on July 22. (ii) Sunday, July 23, is designated a holiday. (iii) Biochemistry concludes the schedule on July 28. (iv) Surgery is held on the day directly following the holiday. (v) Exactly one subject is scheduled between Anatomy and Orthopaedics. Based on this arrangement, which subject is conducted on Wednesday?

A. Biochemistry
B. Physiology
C. Anatomy
D. Microbiology

190 Five siblings were born at regular intervals of 3 years. If the combined sum of their ages equals 60 years, how old is the youngest sibling?

A. 6 years
B. 5 years
C. 4 years
D. 2 years

191 A circular paper sheet is folded and punched with holes. When completely unfolded, what pattern will the sheet reveal?



A. Pattern a
B. Pattern b
C. Pattern c
D. Pattern d

192 Evaluate the relationship between the two statements: (Statement I: Prospective students desiring admission to the BPH program across all colleges must take a single unified entrance examination. Statement II: The Medical Education Commission is the statutory body mandated to organize the centralized common entrance exam for BPH.)

A. Statement I is the cause and Statement II is its effect.
B. Statement II is the cause and Statement I is its effect.
C. Statements I and II represent independent causes and effects.
D. Both statements are mutual effects of an unstated common cause.

193 Indicating a young girl, a boy remarks, "She is the daughter of the son of my uncle's father." How is the girl related to the boy?

A. Sister
B. Daughter-in-law
C. Nephew
D. Mother

194 If directional bearings are rotated such that South-East is reoriented to North and South-West becomes East, what heading does West become?

A. South-East
B. North
C. North-East
D. West

195 Four food containers—vegetables, salads, fruits, and snacks—are arranged vertically in a single column. The top container contains fruits. The fruit container is directly succeeded by snacks, and the vegetable container is positioned between snacks and salads. Which food item is contained in the bottommost box?

A. Snacks
B. Salads
C. Vegetables
D. Fruits

196 A 100-meter straight ladder leans against a vertical building wall, making an angle of 45° with the flat horizontal ground. What is the horizontal distance from the base of the wall to the foot of the ladder?

A. $\frac{100}{\sqrt{2}}$ m
B. $\frac{200}{\sqrt{2}}$ m
C. $100\sqrt{2}$ m
D. $200\sqrt{2}$ m

197 Determine the missing letter denoted by '?' in the fourth row of the table.

44	8	51	F
34	7	56	K
14	5	25	G
33	6	32	?

A. E
B. D
C. F
D. G

198 **Deduce the missing fourth letter in the final four-letter cluster.**

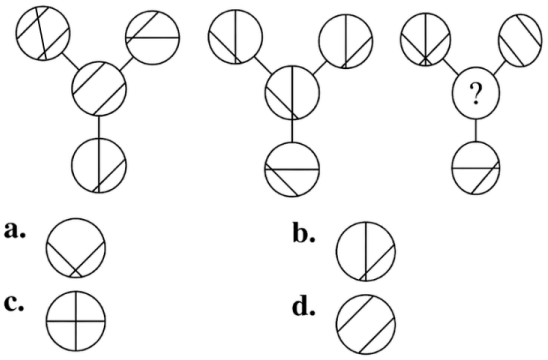
A. M
B. N
C. P
D. S

199 **Find the missing number denoted by '?'_____**

5	17	25
4	15	24
3	8	?

A. 5
B. 6
C. 7
D. 8

200 **Identify which configuration logically advances the transformation sequence.**



A. Configuration a
B. Configuration b
C. Configuration c
D. Configuration d